

Isocolon and comparative construal: A corpus test of rhetorical balance

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Abstract

Rhetorical theory often treats ISOCOLON as more than ornamental balance. A balanced pair of cola can make two units feel more explicitly comparable. A reader-side causal claim would need experimental evidence, but a production-side textual claim can be tested in corpus data. This paper tests whether adjacent discourse-relation pairs in GUM/eRST are more isocolonic when they construe two units as comparable. The outcome is a continuous score combining length balance, syntactic parallelism, and lexical parallelism, rather than a binary figure label. In 17,456 adjacent relation pairs from 299 documents across 24 genres, a broad comparison and co-ordination family shows a clear adjusted increase in isocolonicity, especially for lists, disjunctions, and adversative contrast. A narrow adversative family is smaller, antithesis alone is near zero, and concession is negative. The result supports a restrained textual claim: formal balance is part of the profile of some comparative construals, but it is not a general property of every relation that could be called contrastive.

I INTRODUCTION

Rhetorical theory has long treated figures as patterned forms with argumentative force, not as detachable ornaments. In that tradition, ISOCOLON is interesting because it joins two kinds of evidence that are easy to confuse: formal balance between adjacent units and a reader's sense that the units invite comparison. The motivating claim is simple. When writers place two clauses or phrases in balanced form, they may make the comparison between them more explicit, pointed, or forceful.

This paper asks how much of that claim can be tested without a reader experiment. The answer depends on the explanatory level. At the reader-response level, the claim is causal: isocolon changes what readers notice or how forcefully they construe a relation. Corpus evidence alone can't show that. At the textual level, the claim is distributional: writers may preferentially use balanced form where they construe two units as comparable. That claim is testable.

I therefore test a narrower hypothesis. When a relation places two adjacent units in a comparison-like configuration, the pair should be more isocolonic than pairs expressing cause, condition, elaboration, sequence, or other non-comparative relations. The study uses the GUM corpus and eRST

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annotations because they provide genre-diverse texts, discourse-relation labels, and signal annotation (Zeldes, 2017; Zeldes et al., 2025). The current analysis is preliminary because the RST Signalling Corpus data are still pending, but the design already gives a useful first test.

The results are mixed in the informative sense. Broadly comparative and co-ordinative relations are more formally balanced than matched non-target relations. The effect is clearest for *joint-list*, *joint-disjunction*, and *adversative-contrast*. It is much weaker for a narrow adversative set, near zero for *adversative-antithesis* alone, and negative for *adversative-concession*. The pattern refines the rhetorical claim: isocolonicity tracks some acts of comparative construal better than it tracks the inherited relation name ANTITHESIS.

2 RHETORICAL FIGURES AS FORM-FUNCTION PAIRINGS

The theoretical starting point is Fahnestock’s account of rhetorical figures as form-function pairings. In her work on scientific argument and rhetorical style, figures are not decorative additions to an argument; they are recurrent formal arrangements that can epitomize lines of reasoning (Fahnestock, 1999, 2011). That view matters here because it makes a corpus prediction. If a figure is tied to an argumentative operation, then the figure should be distributed unevenly across the discourse contexts where that operation is being performed.

Fahnestock’s discussion of the figures of argument is especially direct for this project. ISOCOLON concerns approximate equality in the length of adjacent cola, while PARISON concerns similarity of grammatical structure. The two often co-occur, but they are not the same property. A pair can be close in length without being syntactically parallel, and it can be syntactically parallel without being closely balanced in length. The classical tradition links both to comparison and contrast because similar form can make two units cohere as an argumentative set (Fahnestock, 2011).

The same point motivates computational work on rhetorical figures. Harris and Di Marco (2017) argue that figural form is often iconically related to discourse function, so surface patterning can provide evidence about pragmatic and argumentative actions. Harris et al. (2018) develop an annotation scheme for figures in part because figure identification requires attention to both formal regularity and rhetorical role. Lawrence et al. (2017) make the parallel point for argument mining: rhetorical figures can identify regions where argumentative work is being done, but only if formal detection is connected to the content and relation being advanced.

This literature supports the central assumption of the present study. The relevant object is the co-occurrence of surface balance with a discourse relation that makes two units available for comparison. The operational question is whether a graded measure of balance is higher in the relations where rhetorical theory predicts that balance should be useful.

3 FROM CANDIDATE FORM TO RHETORICAL FUNCTION

The main methodological risk is false positives. Repetition and parallelism are common in ordinary text, and only some instances have the rhetorical force that figure theory cares about. Work on automatic chiasmus detection makes the problem vivid. Dubremetz and Nivre (2018) show that trivial repetition patterns are easy to find, but repetitions with rhetorical effect are much harder to separate from accidental or irrelevant patterns. They therefore treat figure detection as a ranking problem rather than a clean binary classification problem.

That lesson carries over to isocolon. Exact repetition, list formatting, dialogue formulas, quoted

slogans, and segmentation artifacts can all look formally balanced. Some are rhetorically relevant, but many are not instances of the target phenomenon. Recent survey work makes the broader point: rhetorical figure detection remains limited by small datasets, inconsistent definitions, and over-reliance on rule-based surface cues (Kühn et al., 2024). A credible corpus test should therefore use surface scores cautiously, validate them against human judgment, and avoid treating high balance as equivalent to a gold figure label.

The analysis below follows that strategy. The isocolonicity score is a continuous predictor-like outcome: it ranks adjacent relation pairs by how much balance they display. The qualitative audit then asks whether high-scoring examples are plausibly isocolonic after excluding exact repetition, dialogue formulas, and other false-positive types. This is a more modest target than automatic figure recognition, but it is the right target for a first rhetorical-function test.

4 CORPUS AND MEASUREMENT

The corpus is the English GUM/eRST dataset. GUM is a genre-diverse corpus with richly annotated English texts (Zeldes, 2017). eRST extends RST-style discourse annotation with graph structures and explicit signal annotation across spoken and written genres (Zeldes et al., 2025). This matters because the project needs discourse relations rather than only adjacent sentences. It also needs enough genre variation to separate a rhetorical effect from a register-specific habit.

The unit of analysis is an adjacent discourse-relation pair. The current dataset contains 17,456 such pairs from 299 documents across 24 genres. For each pair, I computed a composite isocolonicity score from three families of surface evidence. The first is length balance, using token length and related measures of how similar the two units are in size. The second is syntactic parallelism, based on how similar their dependency and part-of-speech profiles are. The third is lexical parallelism, based on repeated or closely corresponding lexical material. The composite should be read as a graded measure of formal balance, rather than as a claim that the pair is an isocolon.

The score is deliberately simple. For any two units with positive length, a balance component is $1 - |n_1 - n_2| / \max(n_1, n_2)$. The length score averages this quantity over word tokens, characters, and estimated syllables. The syntax score averages normalized edit similarity over UPOS, XPOS, and dependency-relation sequences. The lexical score averages lemma-sequence edit similarity and Jaccard overlap over content lemmas. The composite is

$$0.40 \times \text{length} + 0.45 \times \text{syntax} + 0.15 \times \text{lexical}.$$

These weights give syntax slightly more influence than length, with lexical overlap included as a weaker cue.

The target relations were defined in two ways. The broad target family includes *adversative-antithesis*, *adversative-contrast*, *joint-disjunction*, and *joint-list*. This family tests whether formal balance rises in relations that place units in a comparable or co-ordinated frame. The narrow target family includes only *adversative-antithesis* and *adversative-contrast*. Individual-relation analyses then separate the broad effect into its component labels.

The main comparisons adjust for the most obvious structural confounds. Where possible, pairs are compared within documents. The models also include controls for explicit signalling, sentence adjacency, punctuation, mean length, and maximum length. A genre-varying analysis then checks whether the effect is concentrated in one or two genres or survives partial pooling across genres.

The adjusted estimates are ordinary least-squares target coefficients computed by residualizing both the target indicator and the outcome against the controls. The controls are an intercept, explicit relation status, same-sentence adjacency, colon punctuation, semicolon punctuation, standardized log mean length, standardized log maximum length, and document fixed effects. The genre-varying analysis first fits the same adjusted effect within each genre, then uses a normal empirical-Bayes model with genre effects distributed around a shared mean and between-genre standard deviation.

5 VALIDATION

The score behaves as a useful ranking measure, not as a binary detector. Against the small number of eRST *syn-prl* signal labels in the dataset, the composite score has an AUC of 0.731. The syntax subscore is slightly higher at 0.754, the lexical subscore is 0.719, and the length subscore is 0.646 (Table 1). At the 95th percentile, precision is only 3.1%, but recall is 37.0%, a lift of 7.4 over prevalence. At the 97.5th percentile, precision rises to 5.5%, recall is 32.9%, and lift is 13.1. Those numbers are poor for automatic labelling but useful for ranking rare candidates for audit.

Score	AUC	Gold mean	Non-gold mean
Composite	0.731	0.502	0.327
Length	0.646	0.683	0.560
Syntax	0.754	0.441	0.209
Lexical	0.719	0.202	0.057

Table 1: Score validation against the 73 eRST *syn-prl* signal labels in 17,456 adjacent relation pairs.

The qualitative audit is therefore central. Eighty candidate pairs were human-coded, with independent model-assisted reviews used only to surface borderline cases. The combined human coding yielded 33 yes, 43 no, and 4 uncertain judgments. High-scoring broad-target candidates performed well: the first batch had 8 yes judgments out of 10 high broad-but-not-narrow items, and the second had 6 yes judgments out of 6 high broad-target items. By contrast, the restatement/repetition controls had 0 yes judgments out of 9. This contrast is important because exact repetition is the strongest surface correlate of formal balance, while also being a separate false-positive class for the purposes of this study.

The audit also clarified the boundary of the category. Bare identity, apposition, dialogue repair, and exact repetition can be rhetorically relevant without being isocolon. Conversely, weak length balance can still matter when the two cola share a frame strongly enough to invite slot substitution. The coding decisions therefore treat isocolon as a graded rhetorical judgment with false-positive classes rather than a mechanically recoverable surface pattern.

6 PRELIMINARY RESULTS

The broad target family shows a clear adjusted increase in composite isocolonicity. Across all adjacent pairs, the adjusted effect is 0.0415 with a confidence interval from 0.0375 to 0.0455. The subscales show that the effect is driven most strongly by syntax: the adjusted syntax effect is 0.0770, compared with 0.0244 for lexical parallelism and 0.0080 for length balance. That pattern is theoretically useful.

The corpus effect reflects shared formal frames more than similar size. Table 2 summarizes the main adjusted effects and the restatement positive control.

The narrow adversative family is smaller. For *adversative-antithesis* plus *adversative-contrast*, the adjusted composite effect is 0.0130, with a confidence interval from 0.0058 to 0.0202. The individual labels explain the difference. *joint-disjunction* has an adjusted effect of 0.0482, *joint-list* has an adjusted effect of 0.0439, and *adversative-contrast* has an adjusted effect of 0.0241. *adversative-antithesis*, despite its rhetorical name, is near zero at 0.0039, with an interval that crosses zero. *adversative-concession* is negative at -0.0131.

The relation atlas also identifies an important positive control. *restatement-repetition* has the strongest adjusted composite effect, 0.1902. That is exactly what a surface score should find: repeated material is highly balanced. But the human audit shows why this can't be the target effect. Repetition can signal clarification, identity, repair, emphasis, or other rhetorical work; exact repetition remains distinct from balanced comparison. The main analysis therefore treats restatement/repetition as a control and diagnostic, not as evidence for the comparison hypothesis.

Genre-varying estimates point in the same direction. The broad target has a partially pooled mean effect of 0.0390 across 20 genres. The *joint-list* estimate is 0.0398 across 17 genres, and *adversative-contrast* is 0.0291 across the genres where there is enough evidence to estimate it. The narrow adversative estimate is 0.0118. These estimates are not identical across genres, but the broad effect isn't confined to a single genre.

Target	Target n	Target mean	Adjusted effect [95% CI]
Broad family	2,462	0.401	0.041 [0.038, 0.045]
Narrow adversative	605	0.382	0.013 [0.006, 0.020]
<i>joint-list</i>	1,690	0.407	0.044 [0.039, 0.049]
<i>joint-disjunction</i>	167	0.409	0.048 [0.035, 0.062]
<i>adversative-contrast</i>	262	0.408	0.024 [0.013, 0.035]
<i>adversative-antithesis</i>	343	0.363	0.004 [-0.006, 0.013]
<i>adversative-concession</i>	679	0.342	-0.013 [-0.020, -0.006]
<i>restatement-repetition</i>	321	0.580	0.190 [0.181, 0.200]

Table 2: Selected adjusted effects on composite isocolonicity. Estimates use document fixed effects and controls for explicitness, sentence adjacency, punctuation, and length. The restatement row is a positive control rather than a comparison target.

7 INTERPRETATION

The findings support a restricted version of the rhetorical-function claim. Formal balance is elevated where writers construe adjacent units as members of a comparable set, especially in lists, disjunctions, and ordinary adversative contrast. The evidence is weaker for labels that look rhetorically relevant in the taxonomy but don't necessarily require balanced presentation in the text. In particular, *adversative-antithesis* as an eRST relation label only partly overlaps with antithesis in the classical theory of figures.

This distinction matters. If the analysis had simply shown that *adversative-antithesis* was high, the result would have been easy to summarize but theoretically shallow. The actual pattern is more

useful. Isocolonicity appears to be associated with a broader act of placing units into a comparable frame. That frame can be contrastive, but it can also be co-ordinative, disjunctive, or list-like. Conversely, some adversative relations are not especially isocolonic because their function is concession, qualification, or discourse management rather than balanced comparison.

The result also clarifies the scope of the corpus study. The evidence is textual and distributional, rather than psycholinguistic: it concerns where balanced form occurs, not how readers experience those pairs. Nor does a high score by itself establish a rhetorical figure. The result shows instead that a graded measure of balance is systematically higher in particular discourse-relation contexts, and that human audit supports the rhetorical relevance of many high-scoring broad-target cases.

The next robustness step is to add the RST Signalling Corpus once the data are available. Work on discourse signalling has shown that coherence relations are often signalled by more than discourse markers, including lexical, syntactic, graphical, semantic, and genre-based cues. That point is developed in the original signalling annotation work and in later studies of RST signals and multiple signals (Das & Taboada, 2018a, 2018b, 2019; Taboada & Das, 2013). This literature gives the project a stronger comparative baseline. If formal balance behaves like a discourse signal, then it should be testable alongside other signal types, not only as an isolated figure score.

8 CONCLUSION

A corpus study can't prove that isocolon makes readers perceive comparison as more forceful. It can test a cleaner textual claim: writers use formal balance unevenly across discourse relations, and the relations most associated with comparable construals should show more of it. The present results support that claim for a broad family of comparison-like and co-ordinative relations, with the strongest evidence coming from lists, disjunctions, and adversative contrast.

The result gives only limited support to the inherited figure label *ANTITHESIS*. The narrow adversative effect is small, and antithesis as a discourse-relation label is close to zero. That negative finding helps sharpen the theory. Isocolon is a formal resource that writers can use when two units are presented as comparable members of an argumentative or expository set.

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